



Schweizerische Eidgenossenschaft
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Federal Department of the Environment,
Transport, Energy and Communications DETEC
Federal Office for the Environment FOEN
Economics and innovation division

BAFU:20XY

Open-access cross-sectoral life-cycle inventory database

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Who are we - Mandate and role of the FOEN (BAFU)

- The Federal Office for the Environment (FOEN) is the Swiss federal government's centre of excellence for environmental issues
- As such, the FOEN is responsible for preparing and implementing environmental policy and for reporting on the state of the environment
- German translation of “The Federal Office for the Environment (FOEN)”: Bundesamt für Umwelt (BAFU)
- Weblink: [Federal Office for the Environment - Homepage](#)





BAFU:2026 database



Context of the BAFU database

- **Increasing use of Life Cycle Assessment (LCA)** within the Federal administration (e.g. in regulations) and in general is driving higher demand for Life Cycle Inventory (LCI) data.
- The quality of **decisions based on LCA** strongly depends on the **quality** of primary and background data.
- Data **transparency** is a key prerequisite for ensuring the **credibility** of LCA results used within the administration.
- **Data sovereignty** ensures long-term availability of the data, traceability and **reproducibility of the information** used for regulation and policy-making

→ **Transparency, open-access and quality are the pillars of the Swiss Federal Administration's LCI-database BAFU:20XY**



Obligations for an open-access LCI-database

- Legal obligations in Switzerland:
 - Federal [Open Government Data Strategy](#) and law ([EMBAG](#))
 - The [Climate and Innovation Act](#) requires the federal government to assess its carbon footprint and to provide emission factors and other environmental data to companies, cantons, and federal entities for the establishment of net-zero roadmaps
 - According to the 2024–2027 Action Plan for the [Sust. Development Strategy 2030](#), LCIs mandated by the federal administration must be freely accessible as digital datasets

→ Open-access LCI database is not optional — it is mandatory



Why Develop Public body-Coordinated LCA Database?

Objective: Provide a trustworthy and openly accessible data foundation for environmental assessments

- Neutral and credible source of environmental data
- Harmonized methodology and improved comparability
 - fixed generic LCI data ensure that differences in results reflect actual environmental performance (e.g. in public procurement)
- Supports policymaking, business decision-making, and research
- Free access reduces costs and barriers to entry
- Increases transparency across value chains

Key message: Public body coordination creates a common and reliable data foundation for the market and regulatory obligations



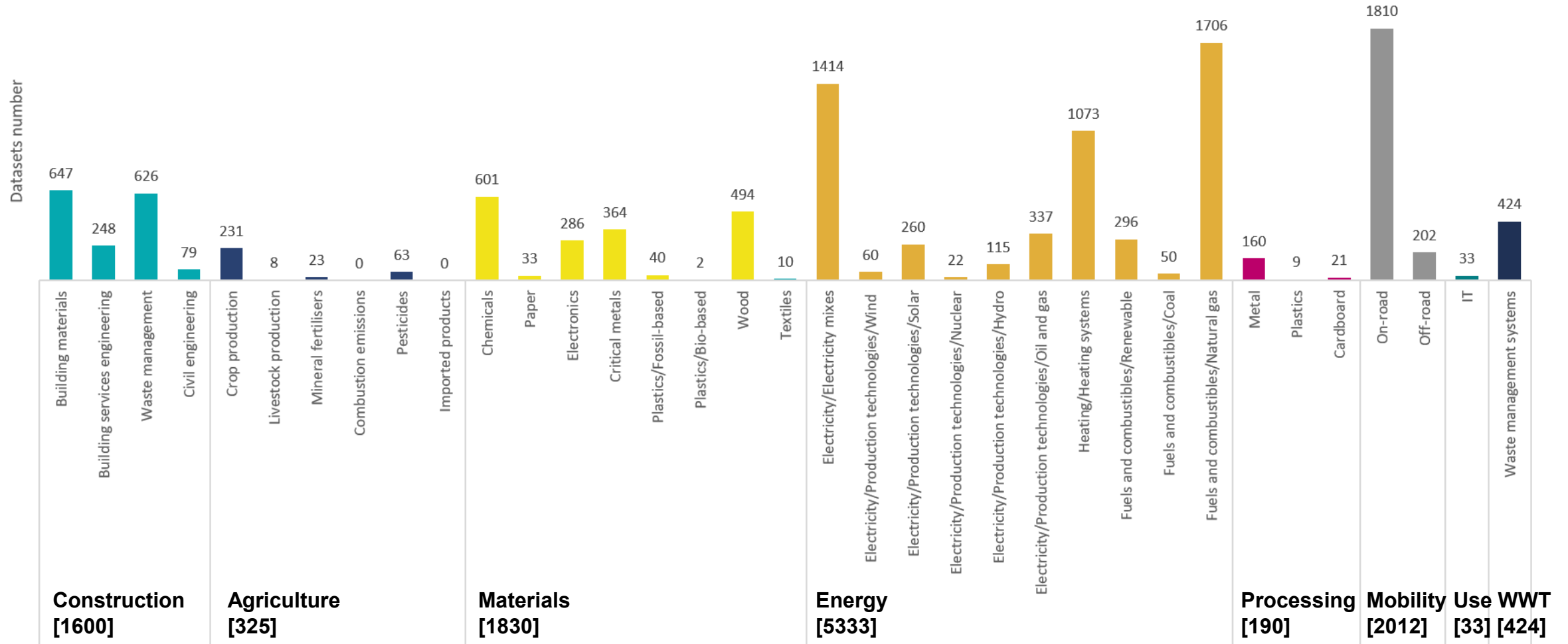
Database release(s)

- **Release date:** 17 December 2025, Mai 2026, ...
- **Content:** 11'747 LCIs (increasing)
- **Description and FAQ:** www.bafu.admin.ch/life-cycle-assessment
- **Scope:** Focus on Switzerland, but many supply chains are similar in other countries + future contribution of partners will enhance the scope as well
- **Objective:** Make available up to date, transparent and comparable datasets without restriction of use
- **Additional information:**
 - Available in ecoSpold v1, Simapro and in zolca formats (compatible with most of the software).
 - Compatibility with the ecological scarcity method (UBP method) and the environmental footprint method (EF) for the impact assessment.
 - The LCIs are described in **reports** which are also available with the datasets or via the [FOEN website](http://FOEN.ch)

→ **Key word: Transparency**



Sectoral coverage in BAFU:2026 (status Mai 2026)



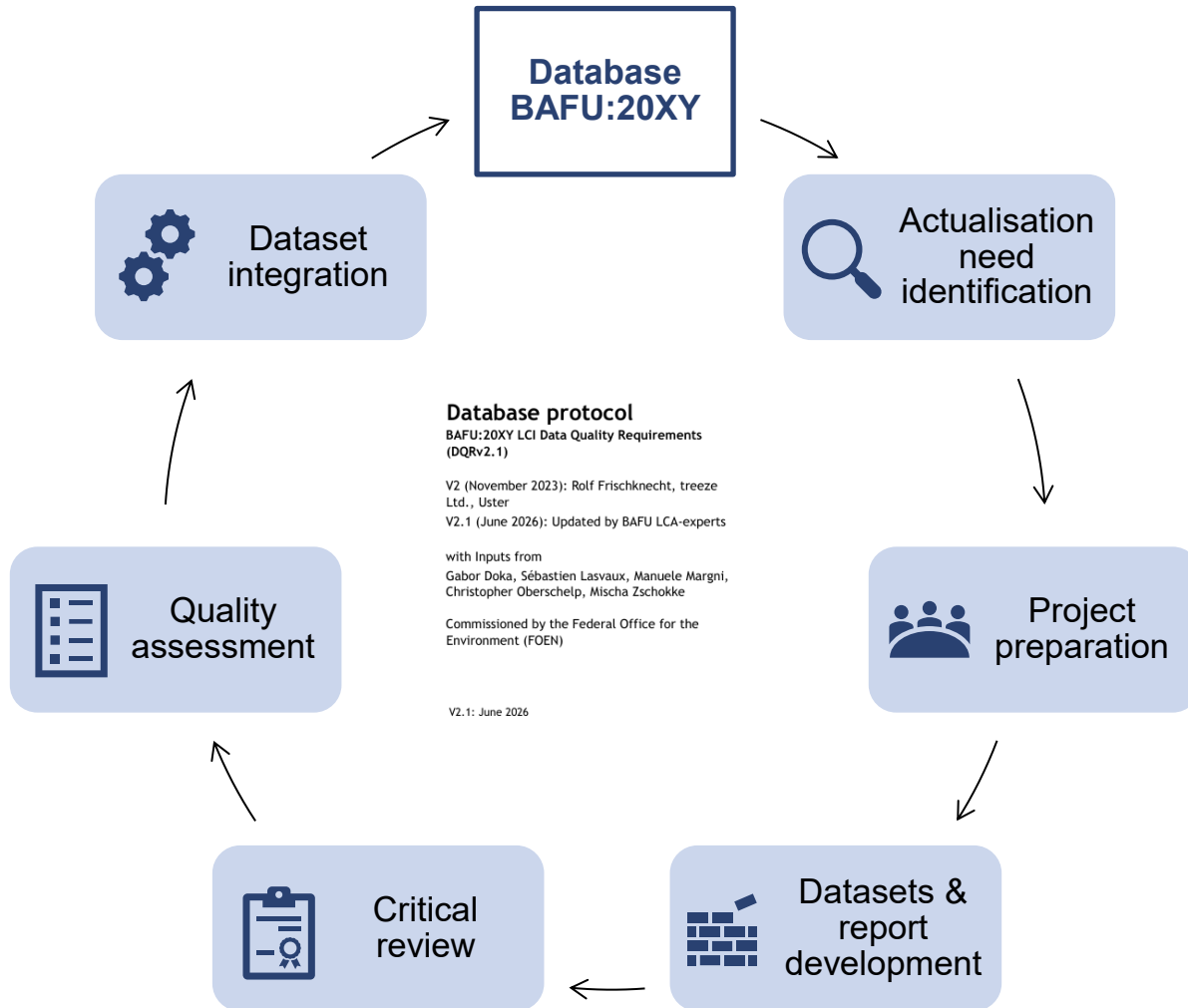


Terms of use

- **Aim:**
 - Ensure the database remains open-access
- **User responsibilities:**
 - Terms of use have to be accepted before accessing the database
 - Non-commercial and commercial use are allowed
 - Commercialisation of the data is forbidden
 - Any changes on BAFU-data have to be transparently described
 - Reports are subjected to IP rights
 - Citation: Life Cycle Inventory database of the Swiss Federal Administration, BAFU:20XX
- **Disclaimer:**
 - No warranty on the models or data



Database development process



- **Consistency across the database is key**
- **Core document is the DQR as it defines the rules for Models, Data integration, Quality assessment.**
Third party review
- **Good input data are essential** → need to set a proper framework for data collection and to ensure their transparency
- **Quality assessment / validation processes are central** for the database development

→ **The LCA community is central to supporting us in achieving the development we aim for**

→ **We are only at the early stage so far...**



Five areas where BAFU:20XY could grow through collaboration

- **Data** : improving existing datasets, adding new ones, develop framework for data collection
- **IT infrastructure** : workflows, versioning, issue tracking, contribution tools
- **Method, Mappings & IT**: Adapted nomenclatures, interoperability, unallocated database, data format
- **Trusted contributions** : transparency, reproducibility, review, formats and quality assurance
- **Governance** : Terms of uses, contribution rules, roles of the community and of other public bodies

→ **Collecting your views on these points is important**



Five areas where BAFU:20XY could grow through collaboration - Concrete examples

We can do it... but not without you!



- **Data** : Keep data up to date, with more frequent updates for dynamic sectors (electricity, etc.), and turn industry support into primary data contributions
- **IT infrastructure** : Development of a framework to ease the data management and the collaborative work
- **Method, Mappings & IT**: develop unallocated dataset and a clear framework to describe system models
- **Trusted contributions** : organize the collaboration, ensure data quality and support of the stakeholders (LCA experts, public bodies, industries, etc.)
- **Governance** : transparent and well defined roles

→ Resource is also a factor for our activities here. Supports from the community and from other public bodies is decisive



Summary: our views for a successful open-access LCI-database

- Transparent and consistent methodology (as little product specific rules, as they are obstacles for comparisons across product groups and modelling e.g. building based on construction materials)
- Clear governance, roles and responsibilities
- Data security
- Independent expert review and quality assurance
- **Collaboration of the community**
- Regular and transparent dataset updates
- Continuous stakeholder engagement
- Long-term public stewardship and secure long term access to data

→ Trust in the database depends as much on how it is governed (DQR, roles of the various actors, quality control processes) as on the quality of the data itself



Conclusion / Key messages



The background **LCIs are at the core of any LCA** and should be transparent and independent from specific industry sectors



The BAFU database will continue its development with other partners to strengthen its development and support across the community

Foreseen development:



Yearly updates of the database are foreseen depending on the needs and new data available (maybe more regular at the beginning to fix bugs)



Our goal is to **further develop** the database in a **collaborative** way



Exchange and coordination with other interested **stakeholders** (national agencies, associations, etc.) in order to set up projects together

→ **Key word: Community support for developing open, high-quality data to enable robust LCAs and informed decision-making**



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Thank you for your attention!

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